Project: Worked examples
Title: Column base plate (EN 1993-1-8)
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## 1. Scope and method

Calculates the bearing, bolt tension, T-stub and shear utilisations of a base plate under an I/H, RHS or CHS column on grout, and the plate thickness that satisfies them. The anchorage is designed outside this sheet.

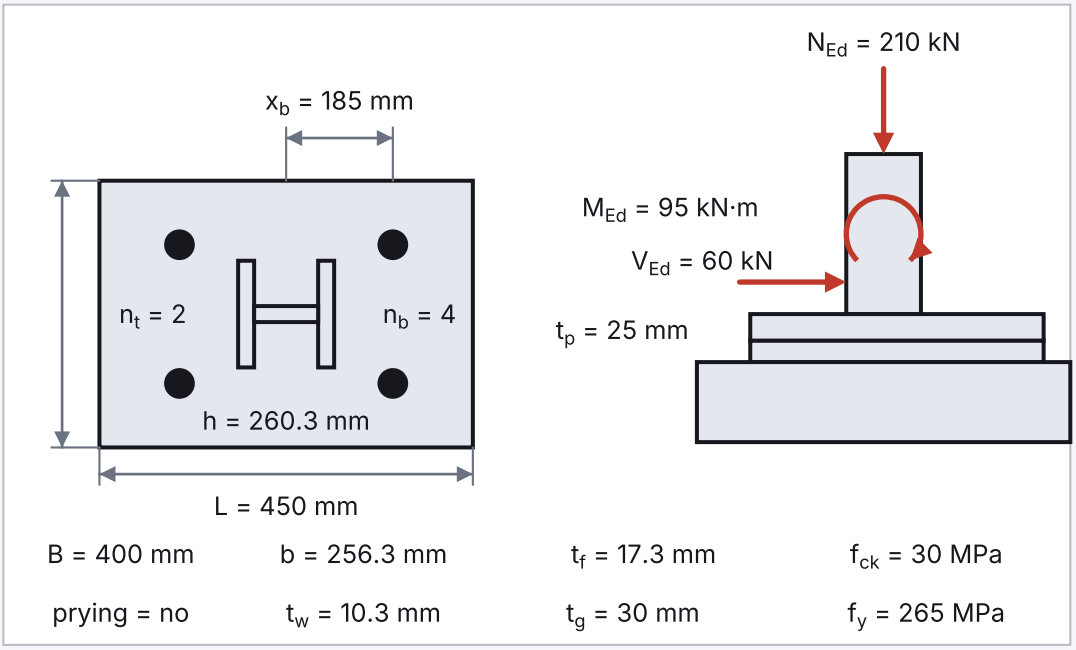



Figure 1 — A base plate B by L in plan under an I/H column, one holding-down bolt row each side at $x_b$ from the axis; in elevation the column on the plate, grout and concrete under $N_{Ed}$, $M_{Ed}$ and $V_{Ed}$. Not to scale; the labels are the inputs as entered.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Bearing (cl. 6.2.5) | $f_{cd} = \alpha_{cc} \cdot f_{ck} / \gamma_c$; $f_{jd} = \beta_j \cdot \alpha \cdot f_{cd}$, $\beta_j = 2/3$; $c = t_p \cdot \sqrt{[f_y / (3 \cdot f_{jd} \cdot \gamma_{M0})]}$ |
| Axial (cl. 6.2.8.2) | $N_{j,Rd} = f_{jd} \cdot A_{eff}$, $A_{eff}$ the bearing strips of width c round the column footprint (full rectangle when they overlap); $u_N = N_{c,Ed} / N_{j,Rd}$ |
| Moment (cl. 6.2.8.3, Table 6.7) | $z_C$ from the section ($(h - t_f)/2$, $(h - t)/2$, half-annulus centroid) or entered, $x_b$ the bolt row from the axis, $z = z_C + x_b$; $C_{Ed} = (M_{Ed} + N_{Ed} \cdot x_b)/z$ and $T_{Ed} = C_{Ed} - N_{Ed}$ when both are ≥ 0; with $N_{Ed} > 0$ and $T_{Ed} < 0$ both sides are in compression, $C = N_{Ed}/2 \pm M_{Ed}/(2z_C)$; with $N_{Ed} < 0$ and $C_{Ed} < 0$ both rows are in tension, $T = -N_{Ed}/2 \pm M_{Ed}/(2x_b)$; $F_{t,Ed} = T_{Ed}/n_t$ at the more loaded row |
| Plate T-stub (cl. 6.2.6.11, 6.2.4, Table 6.2) | $M_{pl,Rd} = 0.25 \cdot l_{eff} \cdot t_p^2 \cdot f_y / \gamma_{M0}$; without prying $\min(2 \cdot M_{pl,Rd}/m, n_t \cdot F_{t,Rd})$; with prying the modes $4 \cdot M_{pl,Rd}/m$, $(2 \cdot M_{pl,Rd} + n \cdot \Sigma F_{t,Rd})/(m + n)$ with $n = \min(e_{min}, 1.25m)$, and $n_t \cdot F_{t,Rd}$; $l_{eff}$ is the total for the $n_t$ bolts |

| WHAT IT CALCULATES | RELATION |
|---|---|
| Compression T-stub (cl. 6.2.6.9, 6.2.6.10) | $C_{Rd} = f_{jd} \cdot A_{flange}$, checked when $M_{Ed} > 0$ and the resultant is in compression |
| Shear (cl. 6.2.2) | $F_{2,vb,Rd} = \alpha_{bc} \cdot f_{ub} \cdot A_s / \gamma_{M2}$, $\alpha_{bc} = 0.44 - 0.0003 \cdot f_{yb}$; $F_{vb,Rd} = \min(F_{1,vb,Rd}, F_{2,vb,Rd})$; $V_{Rd} = C_{f,d} \cdot N_{c,Ed} + n_b \cdot F_{vb,Rd}$; $V_{b,Ed} = \max(V_{Ed} - C_{f,d} \cdot N_{c,Ed}, 0)/n_b$ |
| Required thickness | the smallest $t_p$ in 10–100 mm with $u_N$, $u_C$ and $u_T$ all ≤ 1, $z_C$, $C_{Ed}$ and $T_{Ed}$ re-evaluated at every trial $t_p$ |

Basis: Design resistance of column bases in shear[1](cl. 6.2.2); Equivalent T-stub in tension[1](cl. 6.2.4, Table 6.2); Equivalent T-stub in compression[1](cl. 6.2.5); Concrete in compression including grout[1](cl. 6.2.6.9); Base plate in bending under compression[1](cl. 6.2.6.10); Base plate in bending under tension[1](cl. 6.2.6.11); Anchor bolts in tension[1](cl. 6.2.6.12); Column bases subject to axial forces[1](cl. 6.2.8.2); Column bases subject to axial force and bending moment[1](cl. 6.2.8.3, Table 6.7); Design strength of concrete; partial area loading[2](cl. 3.1.6, 6.7).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| column | Column section type | I/H | | drawing: PF-GA-001 rev C | any |
| prying | Prying forces develop (short bolts, $L_b \le L_b^*$); off is conservative for the plate | no | | assumption: Cast-in holding-down bolts: prying not assumed (min(2·M_pl/m, ΣF_t) is the conservative case). | any |
| B | Plate width | 400 | mm | drawing: PF-BP-001 rev B | 150 … 1200 mm |
| L | Plate length | 450 | mm | drawing: PF-BP-001 rev B | 150 … 1200 mm |
| $t_p$ | Plate thickness | 25 | mm | drawing: PF-BP-001 rev B | 10 … 60 mm |
| $n_t$ | Holding-down bolts in each bolt row (one row each side) | 2 | | drawing: PF-BP-001 rev B | 1 … 6 |
| $n_b$ | Holding-down bolts in total (both sides) | 4 | | drawing: PF-BP-001 rev B | 2 … 12 |
| $x_b$ | Bolt row distance from the column axis (outside the column face, inside the plate edge) | 185 | mm | drawing: PF-BP-001 rev B | 80 … 600 mm |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| m | Bolt to flange-weld toe distance, tension T-stub (required: it follows from $x_b$, the column and the weld) | 46.9 | mm | assumption: Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm. | 10 … 100 mm |
| $l_{eff}$ | Effective length of the tension T-stub, total for the $n_t$ bolts (required: from the cl. 6.2.6.5 patterns for the plate) | 350 | mm | assumption: Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate). | 50 … 800 mm |
| h | Column depth, along the plate length L (I/H and RHS) | 260.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 100 … 1000 mm |
| b | Column width, along the plate width B (I/H and RHS) | 256.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 100 … 1000 mm |
| $t_f$ | Flange thickness (I/H) | 17.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 5 … 50 mm |
| $t_w$ | Web thickness (I/H) | 10.3 | mm | spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy | 4 … 40 mm |
| $t_g$ | Grout thickness (optional; blank = the $\beta_j$ = 2/3 condition is not checked) | 30 | mm | drawing: PF-BP-001 rev B | 5 … 100 mm |
| $f_y$ | Plate yield strength | 265 | MPa | spec: BS EN 10025-2 S275, 16 mm < t ≤ 40 mm | 235 … 440 MPa |
| $f_{ck}$ | Concrete characteristic cylinder strength | 30 | MPa | spec: BS EN 206 C30/37 pad foundation, drawing PF-FN-001 | 20 … 45 MPa |
| $F_{t,Rd}$ | Tension resistance per bolt (from the connection sheet) | 203 | kN | reference: S4 bolt sheet, M24 8.8 (203.04 kN), rounded down | 10 … 1000 kN |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $f_{ub}$ | Anchor bolt ultimate tensile strength | 800 | MPa | spec: EN ISO 898-1 class 8.8 | 400 … 900 MPa |
| $f_{yb}$ | Anchor bolt yield strength ($\alpha_{bc}$ is valid for 235–640 MPa) | 640 | MPa | spec: EN ISO 898-1 class 8.8 | 240 … 640 MPa |
| $A_s$ | Anchor bolt tensile stress area (from the connection sheet) | 352.5 | mm² | reference: S4 bolt sheet, M24 stress area (352.50 mm²) | 80 … 1000 mm² |
| $F_{1,vb,Rd}$ | Bolt bearing resistance on the plate (from the bearing resistance of the connection sheet) | 364.4 | kN | reference: S4 bolt sheet, M24 bearing on 25 mm S275 plate, $e_1$ = 60 mm, $d_o$ = 27 mm (364.44 kN) | 10 … 1000 kN |
| $f_{ck,g}$ | Grout characteristic strength (optional; blank = the $\beta_j$ = 2/3 condition is not checked) | 60 | MPa | spec: Non-shrink grout datasheet, 28-day strength (placeholder GR-DS-01) | 20 … 100 MPa |
| $N_{Ed}$ | Axial force, compression positive and opposite in sign to the member sheet's axial force (a frame reaction $R1_y$ is compression positive) | 210 | kN | reference: S5 load combinations sheet, ULS 6.10b, wind leading, column base A | -2000 … 5000 kN |
| $M_{Ed}$ | Major-axis moment (uniaxial) | 95 | kN·m | reference: S5 load combinations sheet / S2 frame sheet, column base A | 0 … 1000 kN·m |
| $V_{Ed}$ | Shear force | 60 | kN | reference: S5 load combinations sheet / S2 frame sheet, column base A | 0 … 1000 kN |
| $\alpha_{cc}$ | Coefficient for long-term effects on the concrete strength | 0.85 | | spec: UK NA to BS EN 1992-1-1, NA to cl. 3.1.6(1)P | 0.8 … 0.9 |
| $\gamma_c$ | Partial factor for concrete | 1.5 | | spec: UK NA to BS EN 1992-1-1, Table NA.1 | 1.2 … 1.6 |
| $\gamma_{M0}$ | Partial factor for the plate | 1 | | spec: UK NA to BS EN 1993-1-1, NA.2.15 | 1 … 1.15 |
| $\gamma_{M2}$ | Partial factor for the anchor bolts | 1.25 | | spec: UK NA to BS EN 1993-1-8, NA.2.3 | 1.1 … 1.4 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $C_{fd}$ | Friction coefficient, plate to grout | 0.2 | | spec: EN 1993-1-8 cl. 6.2.2, sand-cement grout | 0 … 0.4 |
| α | Concentration factor √($A_{c1}$/$A_{c0}$) of the foundation (1 when the foundation geometry is not known) | 1.5 | | assumption: Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_c1/A_c0 well above 2.25; checked on drawing PF-FN-001. | 1 … 2.5 |
| $\alpha_{basis}$ | Foundation geometry that the concentration factor rests on (state it when the factor is above 1) | Pad 1.2 × 1.2 m, 0.8 m deep (PF-FN-001): A_c1/A_c0 above 2.25, so α = 1.5 is taken | | drawing: PF-FN-001 rev A | any |
| $profis_{ref}$ | Hilti PROFIS report reference (the report is filed as Evidence) | PROFIS-BP1 (to be filed) | | assumption: The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked. | any |

### 3. Assumptions

A1. The axial force is positive in compression, the opposite of the member check sheet. (Sign convention.)
A2. A link from the frame sheet takes the reaction $R1_y$, never the member force $N_{e1}$. ($R1_y$ is up positive, so compression into the base.)
A3. Loading is static, with a major-axis moment only. (Scope of this sheet.)
A4. The plate is unstiffened and rectangular, on grout adequate for $\beta_j$ = 2/3. (The grout is checked when its strength and thickness are entered.)
A5. The simplified T-stub model has one compression resultant and one tension resultant per bolt row. (BS EN 1993-1-8:2005 cl. 6.2.8.3.)
A6. The compression resultant acts at the flange centre, or at the half-annulus centroid of a CHS. (BS EN 1993-1-8:2005 cl. 6.2.8.3.)
A7. The force regime follows Table 6.7, and the web is not credited in the moment check. (BS EN 1993-1-8:2005 cl. 6.2.8.3.)
A8. The effective length $l_{eff}$ is the total for the $n_t$ bolts of a row. (Read from the cl. 6.2.6.5 patterns.)
A9. One $l_{eff}$ serves modes 1 and 2, which is conservative for mode 2. ($l_{eff,1}$ is not more than $l_{eff,2}$.)
A10. The concrete, pull-out, breakout, edge and interaction failures of the anchors are designed in Hilti PROFIS. (From $F_{t,Ed}$ and $V_{b,Ed}$ per anchor; the report is filed as Evidence and referenced.)
A11. With prying switched on, the prying force Q is not in $F_{t,Ed}$ and PROFIS must add it. (External anchorage design.)
A12. The friction resistance counts only the compression, $C_{f,d}$·max($N_{Ed}$, 0). (BS EN 1993-1-8:2005 cl. 6.2.2.)
A13. A column in net tension has no friction resistance. (BS EN 1993-1-8:2005 cl. 6.2.2.)

A14. Anchor bolts share the shear equally. (BS EN 1993-1-8:2005 cl. 6.2.2.)
A15. Anchor bolt holes are no larger than the standard clearance. (Oversized or slotted holes are outside this sheet.)
A16. The bearing outstand c is cut back where it would pass the plate edge. (Conservative against an uncut c.)
A17. The web and the column interior keep the full outstand c. (Conservative against an uncut c.)
A18. The CHS compression side is half the annular bearing area, with its centroid as lever arm. (An approximation.)
A19. The compression T-stub is checked against the flange area alone when $M_{Ed}$ is above 0. (Conservative limitation.)
A20. The whole footprint resists at $M_{Ed}$ = 0, so the compression check jumps just above 0, conservatively. (Limitation.)
A21. Partial factors and the friction coefficient come from the records in use. ($\gamma_c$ and $\alpha_{cc}$: EN 1992-1-1 NA; $\gamma_{M2}$: EN 1993-1-8 NA; $\gamma_{M0}$: EN 1993-1-1 NA; $C_{f,d}$: the grout used.)
A22. The defaults are the EN recommended and UK values. (A small set of factors only; no table from a standard is shipped.)
A23. prying (Prying forces develop (short bolts, $L_b \le L_b^*$); off is conservative for the plate) is taken as no. (Cast-in holding-down bolts: prying not assumed ($\min(2 \cdot M_{pl}/m, \Sigma F_t)$ is the conservative case).)
A24. m (Bolt to flange-weld toe distance, tension T-stub (required: it follows from $x_b$, the column and the weld)) is taken as 46.9 mm. (Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less $0.8 \cdot \sqrt{2}$ × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.)
A25. $l_{eff}$ (Effective length of the tension T-stub, total for the $n_t$ bolts (required: from the cl. 6.2.6.5 patterns for the plate)) is taken as 350 mm. (Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).)
A26. α (Concentration factor $\sqrt{(A_{c1}/A_{c0})}$ of the foundation (1 when the foundation geometry is not known)) is taken as 1.5. (Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives $A_{c1}/A_{c0}$ well above 2.25; checked on drawing PF-FN-001.)
A27. $profis_{ref}$ (Hilti PROFIS report reference (the report is filed as Evidence)) is taken as PROFIS-BP1 (to be filed). (The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked.)

## 4. Working

[31] $f_{cd} = \alpha_{cc} * f_{ck} / \gamma_c$ [2](cl. 3.1.6)
= 0.85 × 30 MPa / 1.5
= 17 MPa
Concrete design compressive strength

[32] $\beta_j$ = 2/3, fixed [1](cl. 6.2.5)
= grout conditions of cl. 6.2.5 assumed met
= 0.66667
Joint coefficient $\beta_j$

[33] $f_{jd} = \beta_j * \alpha * f_{cd}$ [1](cl. 6.2.5)
= 0.66667 × 1.5 × 17 MPa
= 17 MPa
Design bearing strength of the joint

[34] $c = t_p * \mathrm{sqrt}(f_y / (3 * f_{jd} * \gamma_{M0}))$ [1](cl. 6.2.5)
= 25 mm × sqrt(265 MPa / (3 × 17 MPa × 1))
= 56.987 mm
Additional bearing width round the column footprint

[35] $A_{eff} = 2(b + 2c)(t_f + 2c) + (h - 2t_f - 2c)(t_w + 2c)$ [1](cl. 6.2.5)
= 2 × 370.27 × 131.27 + 111.73 × 124.27 mm²
= 111100 mm²
Effective bearing area

[38] $N_{j,Rd} = f_{jd} * A_{eff}$ [1](cl. 6.2.8.2, 6.2.5)
= 17 MPa × 111100 mm²
= 1888.7 kN
Axial bearing resistance of the joint
[39] $N_{c,Ed} = \max(N_{Ed}, 0)$
= $N_{Ed}$ = 210 kN
= 210 kN
Design compression in the column
[40] $u_N = N_{c,Ed} / N_{j,Rd}$
= 210 kN / 1888.7 kN
= 0.11119
Axial bearing utilisation
[41] $t_{req,N}$ = smallest $t_p$ in 10–100 mm with $f_{jd} \cdot A_{eff}(c(t_p)) \geq N_{c,Ed}$ [1](cl. 6.2.5)
(bisection)
= $N_{c,Ed}$ = 210 kN; carried at the 10 mm minimum
= 10 mm
Plate thickness required for axial bearing alone
[42] $z_C = (h - t_f)/2$ [1](cl. 6.2.8.3)
= (260.3 - 17.3)/2 mm
= 121.5 mm
Compression resultant distance from the column axis
[43] $z = z_C + x_b$ [1](cl. 6.2.8.3)
= 121.5 + 185
= 306.5 mm
Lever arm between the compression resultant and the bolt row
[44] $C_{Ed} = (M_{Ed} + N_{Ed} \cdot x_b)/z$ [1](cl. 6.2.8.3, Table 6.7)
= (95 kN·m + 210 kN × 185 mm)/306.5 mm
= 436.7 kN
Compression resultant under the compression flange
[45] $T_{Ed} = C_{Ed} - N_{Ed}$ [1](cl. 6.2.8.3, Table 6.7)
= 436.7 - 210
= 226.7 kN
Tension resultant at the bolt row
[46] $C_{Ed,2}$ = none
= other side
= 0 kN
Compression resultant under the other flange
[47] $T_{Ed,2}$ = none
= other row
= 0 kN
Tension resultant at the other bolt row
[48] $F_{t,Ed} = T_{Ed} / n_t$ [1](cl. 6.2.6.12)
= 226.7 kN / 2
= 113.35 kN
Holding-down bolt tension per bolt at the more loaded row (→ PROFIS)
[49] $A_{flange}$ = (b + 2c)($t_f$ + 2c), at most $A_{eff}/2$ [1](cl. 6.2.6.9)
= min(370.27 × 131.27, $A_{eff}/2$) mm²
= 48608 mm²
Bearing area of the compression flange
[50] $C_{Rd} = f_{jd} * A_{flange}$ [1](cl. 6.2.6.9, 6.2.6.10)
= 17 MPa × 48608 mm²
= 826.33 kN
Bearing resistance of the compression flange

[51] $u_C = C_{Ed} / C_{Rd}$
$= 436.7\ kN / 826.33\ kN$
$= 0.52849$
Compression flange utilisation
[52] $M_{pl,Rd} = 0.25 * l_{eff} * t_p^2 * f_y / \gamma_{M0}$ [1](Table 6.2)
$= 0.25 \times 350\ mm \times (25\ mm)^2 \times 265\ MPa / 1$
$= 14.492\ kN·m$
Plastic moment resistance of the plate over $l_{eff}$
[53] $T_{Rd,1} = 2 * M_{pl,Rd} / m$ [1](Table 6.2)
$= 2 \times 14.492\ kN·m / 46.9\ mm$
$= 618\ kN$
Mode 1-2 without prying forces
[54] $T_{Rd,3} = n_t * F_{t,Rd}$ [1](Table 6.2)
$= 2 \times 203\ kN$
$= 406\ kN$
Mode 3: bolt failure
[55] $T_{Rd} = \min(T_{Rd,1}, T_{Rd,3})$ [1](cl. 6.2.4)
$= \min(618\ kN, 406\ kN)$
$= 406\ kN$
Tension T-stub resistance (no prying)
[56] $u_T = T_{Ed} / T_{Rd}$
$= 226.7\ kN / 406\ kN$
$= 0.55839$
Plate bending and bolts in tension, T-stub utilisation
[57] $u_{bt} = F_{t,Ed} / F_{t,Rd}$
$= 113.35\ kN / 203\ kN$
$= 0.55839$
Holding-down bolt tension utilisation
[58] $t_{req}$ = smallest $t_p$ in 10–100 mm with $u_N$, $u_C$, $u_T \le 1$, with $z_C$, $C_{Ed}$ and $T_{Ed}$ recomputed at every trial $t_p$ (scan and bisection) [1](cl. 6.2.5, 6.2.6.9, 6.2.6.11)
$= N_{c,Ed} = 210\ kN$, $C_{Ed} = 436.7\ kN$, $T_{Ed} = 226.7\ kN$
$= 15.142\ mm$
Plate thickness required
[59] $\alpha_{bc} = 0.44 - 0.0003 \cdot f_{yb}$ [1](cl. 6.2.2)
$= 0.44 - 0.0003 \times 640$
$= 0.248$
Anchor bolt shear factor $\alpha_{bc}$
[60] $F_{2,vb,Rd} = \alpha_{bc} * f_{ub} * A_s / \gamma_{M2}$ [1](cl. 6.2.2)
$= 0.248 \times 800\ MPa \times 352.5\ mm^2 / 1.25$
$= 55.949\ kN$
Anchor bolt shear resistance $F_{2,vb,Rd}$
[61] $F_{vb,Rd} = \min(F_{1,vb,Rd}, F_{2,vb,Rd})$ [1](cl. 6.2.2)
$= \min(364.4\ kN, 55.949\ kN)$
$= 55.949\ kN$
Anchor bolt shear resistance per bolt (bearing or shear)
[62] $F_{f,Rd} = C_{fd} * N_{c,Ed}$ [1](cl. 6.2.2)
$= 0.2 \times 210\ kN$
$= 42\ kN$
Friction resistance, compression only
[63] $V_{Rd} = F_{f,Rd} + n_b * F_{vb,Rd}$ [1](cl. 6.2.2)
$= 42\ kN + 4 \times 55.949\ kN$
$= 265.8\ kN$
Shear resistance of the base

[64] $u_V = V_{Ed} / V_{Rd}$
$= 60\ kN / 265.8\ kN$
$= 0.22574$
Shear utilisation
[65] $V_{b,Ed} = \max(V_{Ed} - F_{f,Rd}, 0)/n_b$ [1](cl. 6.2.2)
$= \max(60 - 42, 0)/4$
$= 4.5\ kN$
Shear per anchor bolt after friction (→ PROFIS)
[66] $u_{max} = \max(u_N, u_C, u_T, u_{bt}, u_V)$
$= u_N = 0.11119,\ u_C = 0.52849,\ u_T = 0.55839,\ u_{bt} = 0.55839,\ u_V = 0.22574$
$= 0.55839$
Governing utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $F_{t,Ed}$ | 113.35 kN | Bolt tension per bolt (to PROFIS) |
| $V_{b,Ed}$ | 4.500 kN | Shear per anchor bolt (to PROFIS) |
| $N_{j,Rd}$ | 1888.7 kN | Axial bearing resistance |
| $u_{max}$ | 0.56 | Governing utilisation |
| $f_{cd}$ | 17 MPa | Concrete design strength |
| $\beta_j$ | 0.66667 | Joint coefficient $\beta_j$ |
| $f_{jd}$ | 17 MPa | Design bearing strength of the joint |
| c | 56.99 mm | Bearing outstand round the column footprint |
| $A_{eff}$ | 111100 mm² | Effective bearing area |
| $t_{req,N}$ | 10 mm | Plate thickness required for axial bearing alone |
| $t_{req}$ | 15.14 mm | Plate thickness required (bearing, compression and tension T-stubs) |
| $z_C$ | 121.5 mm | Distance from the column axis to the compression resultant |
| z | 306.5 mm | Lever arm between the compression resultant and the bolt row |
| $C_{Ed}$ | 436.70 kN | Compression resultant |
| $C_{Ed,2}$ | 0 kN | Compression resultant, other side |
| $T_{Ed}$ | 226.70 kN | Tension resultant, more loaded row |
| $T_{Ed,2}$ | 0 kN | Tension resultant, other row |
| $A_{flange}$ | 48608 mm² | Compression-side bearing area |
| $C_{Rd}$ | 826.33 kN | Compression-side resistance |
| $M_{pl,Rd}$ | 14.49 kN·m | Plate plastic moment |
| $T_{Rd,1}$ | 618.00 kN | T-stub mode 1-2, no prying |
| $T_{Rd,3}$ | 406 kN | T-stub mode 3 |
| $T_{Rd}$ | 406 kN | Tension T-stub resistance |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $\alpha_{bc}$ | 0.248 | Anchor shear factor $\alpha_{bc}$ |
| $F_{2,vb,Rd}$ | 55.95 kN | Anchor shear resistance $F_{2,vb,Rd}$ |
| $F_{vb,Rd}$ | 55.95 kN | Anchor shear resistance per bolt |
| $F_{f,Rd}$ | 42 kN | Friction resistance |
| $V_{Rd}$ | 265.80 kN | Shear resistance |
| regime | "tension at one bolt row, compression under the other flange (Table 6.7, z = $z_c$ + $x_b$)" | Force regime (Table 6.7) |
| $T_{mode}$ | "mode 3 (bolt failure)" | Governing T-stub mode |
| $u_N$ | 0.11 | Axial bearing utilisation |
| $u_C$ | 0.53 | Compression-side utilisation |
| $u_T$ | 0.56 | Tension T-stub utilisation (governing) |
| $u_{bt}$ | 0.56 | Bolt tension utilisation |
| $u_V$ | 0.23 | Shear utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Tension at the bolt row against the plate T-stub resistance | 226.70 kN | 406 kN | 0.56 | [1](cl. 6.2.6.11) |
| PASS | Bolt tension against the bolt tension resistance | 113.35 kN | 203 kN | 0.56 | [1](cl. 6.2.6.12) |
| PASS | Compression force against compression flange bearing resistance | 436.70 kN | 826.33 kN | 0.53 | [1](cl. 6.2.6.9, 6.2.6.10) |
| PASS | Shear force against friction plus anchor bolt shear resistance | 60 kN | 265.80 kN | 0.23 | [1](cl. 6.2.2) |
| PASS | Axial force against concrete bearing resistance | 210 kN | 1888.7 kN | 0.11 | [1](cl. 6.2.8.2, 6.2.5) |

## 7. Applicability

WARNING records: tabulated values not taken from a reference record are $\alpha_{cc}$, $\gamma_c$ (EN 1992-1-1 National Annex factors); $\gamma_{M2}$ (EN 1993-1-8 National Annex factor); $\gamma_{M0}$ (EN

1993-1-1 National Annex factor); $C_{fd}$ (Friction coefficient plate to grout); $f_y$ (Plate strength); $f_{ck}$ (Concrete strength classes). Enter each as one so its source, edition and licence travel with it. [record]
WARNING $f_y$: one strength of 265 MPa is used at every thickness, but EN 10025-2 lowers it above 16 mm. Enter the strength for the thickness in use, because $t_p$ exceeds that step. [$f_{y,thickness}$]
WARNING $u_{profis}$: the anchorage is not assessed here, and $u_{max}$ covers the steel parts only. Enter the PROFIS report's governing utilisation to include the concrete failure modes of the anchors. [anchorage-result]
WARNING $profis_{ref}$: 'PROFIS-BP1 (to be filed)' is typed text, not an attached report. Attach the PROFIS report as Evidence, because the anchorage is not checked here. [anchorage]

## 8. Verification

```
Template struct.base-plate@1.8.0 Content hash a1c272121ecc4f6f Engine calc-engine/1.0.0
Known-answer cases passed 9/9 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.
2. BS EN 1992-1-1:2004.